N. Thirugnanamuthu's MB-Spectral Contraction Theory of AI Computation

A Linear Algebra Framework for Latency, Load, and Speed-Error Equilibrium

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Abstract

We formalise three intuitions about AI computation—more thinking time, more reasoning steps and less haste give more accurate outputs—as three laws of one linear-algebra model, the MB-Spectral Contraction Theory. The error of a reasoning process evolves as $E(\tau) = e^{-\alpha G\tau}E(0)$, with G symmetric positive definite and $\tau = dT$ the product of reasoning depth d and time T per step. We prove: (Law I) the error is strictly decreasing in T, equals $\|E(0)\|$ at $T = 0$ and vanishes as $T \to \infty$; (Law II) a chain of d contractions with factor $\rho < 1$ has error at most $\rho^d\|E(0)\|$ at cost linear in d; (Law III) the error is strictly increasing in speed $v = 1/\tau$, and for a latency price β a unique optimal latency exists. We show that the first-order (Euler) form of the model is stable only for $T < 2/(\alpha g_{\max})$, which corrects the claim that its error vanishes as $T \to \infty$, and derive the optimal step $T^* = 2/\big(\alpha(g_{\min} + g_{\max})\big)$ with contraction $(\kappa-1)/(\kappa+1)$. All claims are checked on 5000 random spectra with zero violations. On eight published accuracy-versus-compute curves (33 points), accuracy is monotone in compute in 8 of 8 and shows diminishing returns within noise in 6 of 8; a single-mode exponential fits sampling curves but is outperformed by a logistic on token-budget curves with an onset threshold, so the theory needs an onset (dead-time) extension.

Index Terms—Spectral contraction, test-time compute, chain of thought, latency–accuracy trade-off, matrix exponential, Richardson iteration.

I. Introduction

Fast answers from an AI system are often less accurate than answers produced after more deliberation. Chain-of-thought prompting [1] and test-time compute scaling [2] make this concrete, and the speed–accuracy trade-off is a long-standing theme in cognitive science [3]. This paper asks whether the qualitative pattern can be captured by a single linear-algebra model whose properties can be *proved*, not only observed.

We state three laws: **Law I** (latency and perfection): more time per step gives smaller error; **Law II** (computational load): more reasoning steps cost proportionally more and shrink the error geometrically; **Law III** (speed–error equilibrium): speed and accuracy trade off, and an optimal compromise exists once latency has a price. Fig. 1 summarises the setting.

Contributions. (i) A spectral contraction model unifying the three laws (Section II–III). (ii) Step-by-step proofs, including a *correction*: in the first-order form of the model the error does *not* vanish as $T \to \infty$; it diverges beyond a stability limit (Proposition 1). (iii) A closed-form optimal step and a unique latency equilibrium (Proposition 1, Theorem 4). (iv) A numerical verification on 5000 random spectra and a dashboard of results (Section VI). (v) A pre-declared test on eight published accuracy-versus-compute curves, with the program output reproduced verbatim (Section VII).

Relation to prior work. Contraction mappings [4], the matrix exponential [7] and Richardson iteration [6] are classical. The contribution here is the formulation of the three laws in this framework, the corrected stability analysis and the verification; we do not claim the laws hold for any particular deployed model. The framework extends the spectral methods of the author's MB-Universal Matrix [8], where the second eigenvalue of a stochastic matrix controls convergence; here the extreme eigenvalues of G control error contraction.

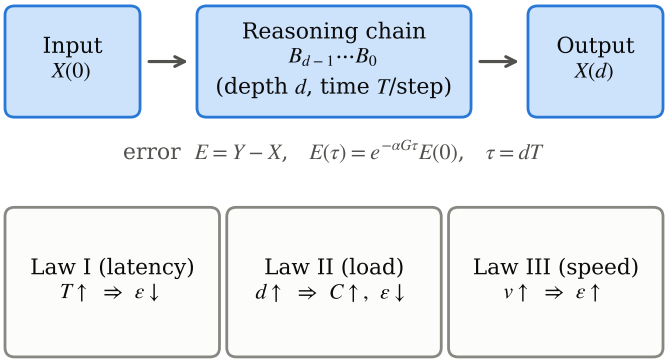


Figure 1. Framework: a reasoning chain of d steps, each of duration T, drives the error $E = Y - X$ through $e^{-\alpha G\tau}$ with $\tau = dT$.

II. Problem Formulation

Let $X(\tau) \in \mathbb{R}^n$ be the internal state of a reasoning process at (computation) time τ, and Y the correct output. The error is $E = Y - X$.

Definition 1 (Relaxation model)**.** Let $M \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues in $(-1, 1)$, $G = I - M$ and $\alpha > 0$. The state

obeys $\dot{X} = \alpha G(Y - X)$, i.e.

$$\dot{E} = -\alpha G E, \qquad E(\tau) = e^{-\alpha G \tau} E(0). \tag{1}$$

G is symmetric positive definite with eigenvalues $0 < g_1 \leq \cdots \leq g_n < 2$, $g_i = 1 - \mu_i$, where μ_i are the eigenvalues of M. Write $\kappa = g_n / g_1$.

Running the process for d steps of duration T gives $\tau = dT$ (semigroup property: $(e^{-\alpha TG})^d = e^{-\alpha dTG}$). The *first-order (Euler) form*, which is the form used in the initial statement of the laws, is

$$A_E(T) = I - \alpha T\, G = (1 - \alpha T) I + \alpha T\, M, \quad E_{t+1} = A_E(T) E_t. \tag{2}$$

Remark 1 (Consistency of the fixed point). A recursion $X_{t+1} = AX_t$ with $Y = AY$ and $\rho(M) < 1$ forces $Y = 0$, since 1 is then not an eigenvalue of A. We therefore use the affine update $X_{t+1} = AX_t + (I - A)Y$, whose fixed point is Y for any A and which gives $E_{t+1} = AE_t$ exactly.

Assumptions. (A1) the relaxation model above; (A2) each reasoning step has cost c_0, so $C(d) = c_0 d$; (A3) latency is $\tau = dT$; speed is $v = 1/\tau$.

III. Spectral Contraction

Theorem 1 (Sandwich bound). *Let $E_0 \neq 0$ and $E(\tau) = e^{-\alpha G \tau} E_0$. For all $\tau \geq 0$:*

$$e^{-\alpha g_n \tau} \|E_0\| \leq \|E(\tau)\| \leq e^{-\alpha g_1 \tau} \|E_0\|, \tag{3}$$

$$\|E(\tau)\| \text{ is strictly decreasing in } \tau, \tag{4}$$

$$\|E(0)\| = \|E_0\|, \qquad \lim_{\tau \to \infty} \|E(\tau)\| = 0. \tag{5}$$

Proof. Step 1. Let $G = \sum_i g_i v_i v_i^\top$ with orthonormal v_i and $E_0 = \sum_i c_i v_i$, so $\sum c_i^2 = \|E_0\|^2$.
Step 2. $E(\tau) = \sum_i c_i e^{-\alpha g_i \tau} v_i$, hence $\|E(\tau)\|^2 = \sum_i c_i^2 e^{-2\alpha g_i \tau}$.
Step 3. Since $g_1 \leq g_i \leq g_n$, $e^{-2\alpha g_n \tau} \leq e^{-2\alpha g_i \tau} \leq e^{-2\alpha g_1 \tau}$. Multiply by c_i^2, sum, and take square roots: (3).
Step 4. $\frac{d}{d\tau} \|E\|^2 = -2\alpha \sum_i g_i c_i^2 e^{-2\alpha g_i \tau} < 0$ because $g_i > 0$ and some $c_i \neq 0$. So $\|E\|$ is strictly decreasing.
Step 5. At $\tau = 0$ the sum is $\sum c_i^2$; as $\tau \to \infty$ the upper bound in (3) tends to 0. ∎

The lower bound shows that perfection is only asymptotic: $\|E(\tau)\| > 0$ for every finite τ.

IV. The Three Laws

A. Law I: latency and perfection

Theorem 2 (Law I). *For a fixed number t of steps, let $\varepsilon_t(T) = \|(e^{-\alpha TG})^t E_0\|$. Then ε_t is strictly decreasing in T, $\lim_{T \to 0} \varepsilon_t = \|E_0\|$ (maximal error: hurry) and $\lim_{T \to \infty} \varepsilon_t = 0$ (perfection: patience).*

Proof. By the semigroup property $\varepsilon_t(T) = \|E(tT)\|$ with $\tau = tT$ increasing in T. Apply Theorem 1: strict decrease, value $\|E_0\|$ at $\tau = 0$ (continuity), limit 0 as $\tau \to \infty$. ∎

Proposition 1 (Euler form: stability limit, correction). *For $A_E(T) = I - \alpha TG$ with eigenvalues $\lambda_i = 1 - \alpha T g_i$:*

(a) $\rho(A_E) < 1 \iff 0 < T < 2/(\alpha g_n)$.

(b) If $T > 2/(\alpha g_n)$ and $c_n \neq 0$ then $\|E_t\| \to \infty$.

(c) $|\lambda_i - e^{-\alpha T g_i}| \leq (\alpha T g_i)^2/2$.

(d) $\rho(A_E)$ *is minimised at* $T^* = \dfrac{2}{\alpha(g_1 + g_n)}$, *with* $\rho^* = \dfrac{\kappa - 1}{\kappa + 1}$.

Proof. (a) $|1 - x_i| < 1 \iff 0 < x_i < 2$ with $x_i = \alpha T g_i$. All i satisfy this iff the largest does: $\alpha T g_n < 2$.
(b) $|\lambda_n| = \alpha T g_n - 1 > 1$ and $\|E_t\| \geq |c_n| |\lambda_n|^t \to \infty$.
(c) By Taylor's theorem $e^{-x} = 1 - x + \frac{x^2}{2} e^{-\xi}$, $\xi \in (0, x)$, so $0 \leq e^{-x} - (1 - x) \leq x^2/2$.
(d) Step 1: $|1 - \alpha T g|$ is convex in g, so $\rho(T) = \max\{|1 - \alpha T g_1|, |1 - \alpha T g_n|\}$.
Step 2: For $T \leq T^*$, $\alpha T g_n - 1 \leq 1 - \alpha T g_1$ (equivalent to $\alpha T(g_1 + g_n) \leq 2$), so $\rho(T) = 1 - \alpha T g_1$, decreasing. For $T \geq T^*$, $\rho(T) = \alpha T g_n - 1$, increasing.
Step 3: Equating $1 - \alpha T g_1 = \alpha T g_n - 1$ gives T^*; then $\rho^* = 1 - \frac{2 g_1}{g_1 + g_n} = \frac{g_n - g_1}{g_n + g_1} = \frac{\kappa - 1}{\kappa + 1}$. ∎

Remark 2. Proposition 1 shows that in the Euler form the error is *not* monotone in T: it falls until about T^*, then rises, and diverges beyond $T = 2/(\alpha g_n)$. The statement "error vanishes as $T \to \infty$" therefore holds for the exact dynamics (Theorem 2) and for the Euler form only while $\alpha T g_n \ll 1$.

B. Law II: computational load

Theorem 3 (Law II). *Let $E_{k+1} = B_k E_k$ for $k = 0, \ldots, d-1$ with $\|B_k\|_2 \leq \rho < 1$, each step costing c_0. Then*

(a) $\|E_{k+1}\| \leq \rho \|E_k\|$ *(strictly decreasing unless $E_k = 0$);*

(b) $\|E_d\| \leq \rho^d \|E_0\|$;

(c) with load $C(d) = c_0 d$: $\|E_d\| \leq \|E_0\| \exp\!\big(-\frac{\ln(1/\rho)}{c_0} C\big)$;

(d) reaching error ε requires $d \geq \ln(\|E_0\|/\varepsilon)/\ln(1/\rho)$.

Proof. Step 1. $\|E_{k+1}\| = \|B_k E_k\| \leq \|B_k\|_2 \|E_k\| \leq \rho \|E_k\|$: (a).
Step 2. Induction on k gives (b).
Step 3. Substitute $d = C/c_0$ in (b) and use $\rho^d = e^{-d \ln(1/\rho)}$: (c).
Step 4. Solve $\rho^d \|E_0\| \leq \varepsilon$ for d: (d). ∎

For the exact dynamics $B_k = e^{-\alpha TG}$ we have $\rho = e^{-\alpha g_1 T}$, so the load needed is $d \geq \ln(\|E_0\|/\varepsilon)/(\alpha g_1 T)$: longer steps reduce the number of steps needed in proportion.

Remark 3. $\rho(B) < 1$ alone does not give $\|B\|_2 < 1$. For $B = \left[\begin{smallmatrix} 0.5 & 5 \\ 0 & 0.5 \end{smallmatrix}\right]$: $\rho(B) = 0.50$, $\|B\|_2 = 5.05$ and $\sup_d \|B^d e_2\| = 5.02 > 1$ (transient growth). For any $\varepsilon > 0$ a norm with $\|B\| \leq \rho(B) + \varepsilon$ exists [5], so the hypothesis means "contractive in a suitable norm".

C. Law III: speed–error equilibrium

Let $\varepsilon(\tau) = \|E(\tau)\|$ and speed $v = 1/\tau$.

Theorem 4 (Law III). *(a) ε as a function of v is strictly increasing: $d\varepsilon/dv > 0$ and $d\varepsilon/d\tau < 0$. (b) $\varepsilon(\tau)$ is strictly convex. (c) For a latency price $\beta > 0$ let $J(\tau) = \varepsilon(\tau) + \beta\tau$. J has a*

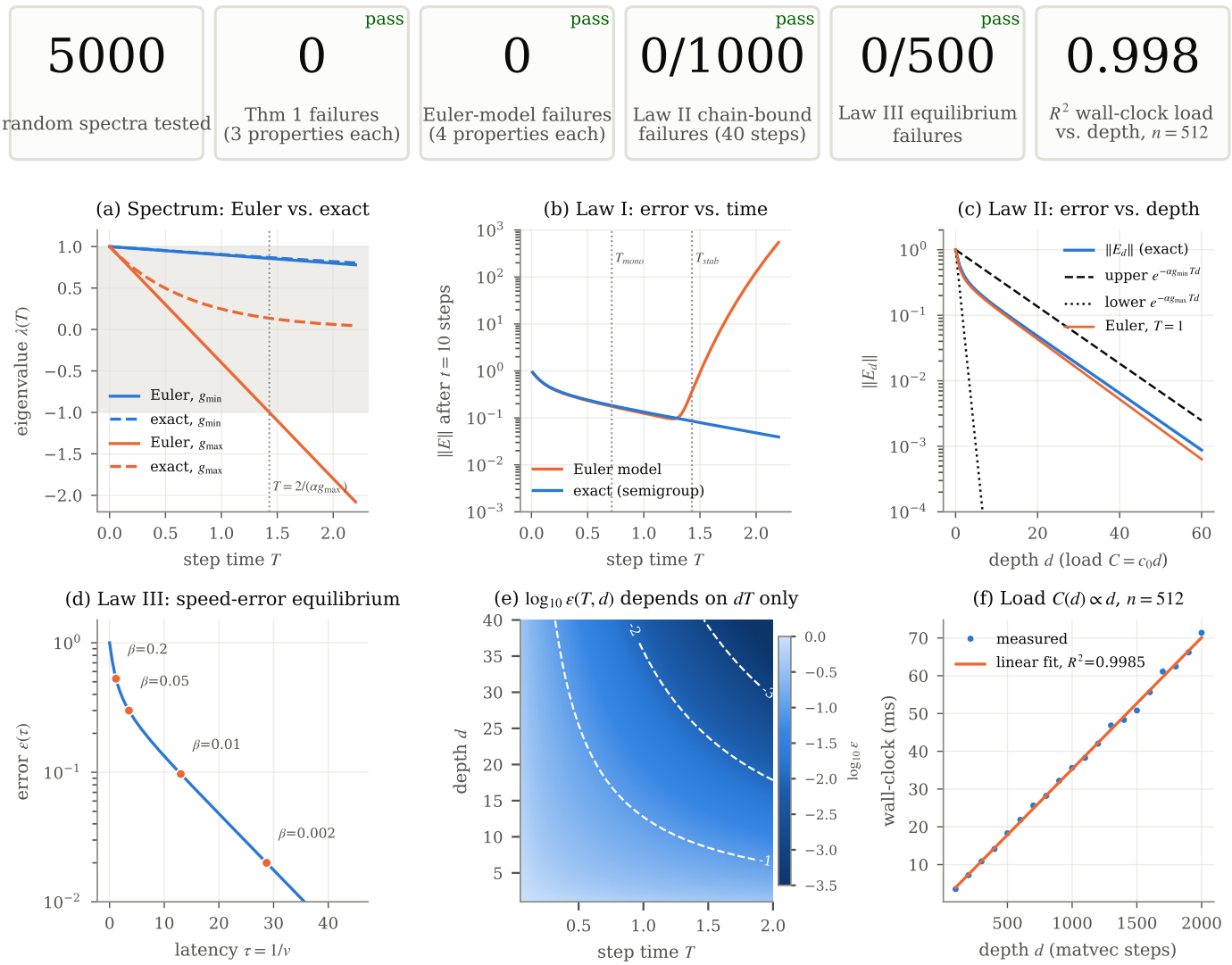


Figure 2. Verification dashboard. Top: pass counts from 5000 random spectra. (a) eigenvalues of the Euler and exact models; (b) Law I: error after 10 steps versus step time; (c) Law II: error versus depth with the Theorem 1 sandwich; (d) Law III: error–latency frontier with equilibria for latency prices β; (e) joint map of $\log_{10} \varepsilon(T, d)$, whose contours are the iso-latency curves $dT = \text{const}$; (f) measured wall-clock load for a 512×512 matrix–vector chain. Main instance: $g_{\min} = 0.10$, $g_{\max} = 1.40$, $\kappa = 14$, $\alpha = 1$, $\|E_0\| = 1$.

unique minimiser τ^*; $\tau^* = 0$ iff $\beta \geq \alpha E_0^\top G E_0 / \|E_0\|$, otherwise $\tau^* > 0$ is the unique root of $-\varepsilon'(\tau) = \beta$. (d) For a single mode ($\varepsilon = \varepsilon_0 e^{-\alpha g \tau}$): $\tau^* = \frac{1}{\alpha g} \ln \frac{\alpha g \varepsilon_0}{\beta}$ and $\varepsilon(\tau^*) = \beta/(\alpha g)$.

Proof. *(a)* $d\varepsilon/d\tau < 0$ by Theorem 1. With $\tau = 1/v$, $d\varepsilon/dv = \varepsilon'(\tau) \cdot (-1/v^2) = -\varepsilon'(\tau)\tau^2 > 0$.
(b) Step 1: Let $w_i = c_i^2 e^{-2\alpha g_i \tau}$, $h = \varepsilon^2 = \sum w_i$. Then $h' = -2\alpha \sum g_i w_i$, $h'' = 4\alpha^2 \sum g_i^2 w_i$.
Step 2: $\varepsilon'' = \dfrac{2hh'' - h'^2}{4h^{3/2}}$ and $2hh'' - h'^2 = 4\alpha^2 \left[2 \sum w_i \sum g_i^2 w_i - (\sum g_i w_i)^2\right]$.
Step 3: Cauchy–Schwarz gives $(\sum g_i w_i)^2 \leq \sum w_i \sum g_i^2 w_i$, so the bracket is $\geq \sum w_i \sum g_i^2 w_i > 0$. Hence $\varepsilon'' > 0$.
(c) Step 1: $J'' = \varepsilon'' > 0$, so J is strictly convex and has at most one minimiser on $[0, \infty)$; $J'(\tau) = \varepsilon'(\tau) + \beta$ increases from $\varepsilon'(0) + \beta$ to $\beta > 0$.
Step 2: $\varepsilon'(0) = -\alpha \sum g_i c_i^2 / \|E_0\| = -\alpha E_0^\top G E_0 / \|E_0\|$. If $\varepsilon'(0) + \beta \geq 0$ then $J' \geq 0$ and $\tau^* = 0$. Otherwise J' changes sign exactly once and τ^* is its root.
(d) $-\varepsilon' = \alpha g \varepsilon_0 e^{-\alpha g \tau} = \beta \Rightarrow e^{-\alpha g \tau^*} = \beta/(\alpha g \varepsilon_0)$; substituting gives both formulas. ∎

Interpretation. At the equilibrium the marginal error reduction per unit latency equals the price of latency. In the single-mode case the residual error at the optimum, $\beta/(\alpha g)$, does not depend on the initial error.

Unified statement. For d steps of time T: $E = e^{-\alpha G T d} E_0$; Law I is monotonicity in T, Law II the bound in d and linear cost, Law III monotonicity in $v = 1/(dT)$ and the equilibrium τ^*. A consequence is the *time–depth equivalence*: the exact error

Table 1. Numerical verification (violations / instances).

Property	Instances	Violations
Thm 1: sandwich bound (3)	5000	0
Thm 1: strict monotone decrease	5000	0
Thm 4(b): convexity of $\varepsilon(\tau)$	5000	0
Prop 1(a): Euler stability limit	5000	0
Prop 1(b): divergence beyond the limit	5000	0
Prop 1(c): $\lvert\lambda_i - e^{-x_i}\rvert \le x_i^2/2$	5000	0
Prop 1(d): optimum T^*, ρ^*	5000	0
Thm 3: $\lVert E_d\rVert \le \rho^d \lVert E_0\rVert$ (1000×40)	1000	0
Thm 3(a): strict decrease at each step	1000	0
Thm 4(c): equilibrium τ^* (interior, boundary)	500	0

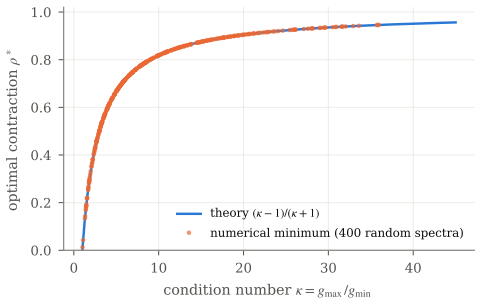


Figure 3. Optimal Euler contraction $\rho^* = (\kappa-1)/(\kappa+1)$ (Proposition 1d): theory versus numerical minimum on 400 random spectra.

depends on T and d only through their product (Fig. 2e).

V. Worked Example (hand calculation)

Take $M = \left[\begin{smallmatrix} 0.5 & 0.2 \\ 0.2 & 0.3 \end{smallmatrix}\right]$, $\alpha = 1$, $E_0 = (1,0)^\top$.

Step 1. $G = I - M = \left[\begin{smallmatrix} 0.5 & -0.2 \\ -0.2 & 0.7 \end{smallmatrix}\right]$, $\operatorname{tr} G = 1.2$, $\det G = 0.35 - 0.04 = 0.31$.

Step 2. $g = \frac{1.2 \pm \sqrt{1.44 - 1.24}}{2}$: $g_1 = 0.3764$, $g_2 = 0.8236$ (both in $(0,2)$).

Step 3. Modal weights $c_1^2 = 0.7236$, $c_2^2 = 0.2764$ (sum $= 1$).

Step 4 (exact). $\lVert E(\tau)\rVert^2 = 0.7236 e^{-2\cdot 0.3764\tau} + 0.2764 e^{-2\cdot 0.8236\tau}$: $\lVert E(1)\rVert = 0.6278$, $\lVert E(3)\rVert = 0.2786$.

Step 5 (Euler, $T = 1$). $\lambda = 1 - g = 0.6236,\ 0.1764$ (exact: $e^{-g} = 0.6863,\ 0.4388$). Errors after 1 and 3 steps: 0.5385, 0.2063, below the exact values because $1 - x \le e^{-x}$.

Step 6 (stability limit). $2/g_2 = 2.428$. For $T = 2.6$: $\lvert 1 - 2.6 g_2\rvert = 1.1414 > 1$; after 10 steps the error is $1.97 > \lVert E_0\rVert = 1$ (divergence).

Step 7 (optimum). $T^* = 2/(g_1 + g_2) = 1.667$, $\rho^* = (g_2 - g_1)/(g_2 + g_1) = 0.3727$.

VI. Numerical Verification

The proofs above are analytic; the numerical tests guard against algebra and implementation slips. We drew 5000 random instances with $n \in [3,12]$, $\mu_i \sim U(-0.9, 0.95)$, $\alpha \sim U(0.2, 2)$ and Gaussian c_i. Table 1 lists the checks. Law II was tested on 1000 chains of 40 random, non-normal matrices with $\lVert B_k\rVert_2 \le 0.9$; Law III on 500 instances (400 interior and 100 boundary prices), comparing $J(\tau^*)$ with a dense grid. The single-mode formula with $\varepsilon_0 = 0.8$, $g = 0.37$, $\beta = 0.05$ gives $\tau^* = 4.806$ analytically and 4.806 numerically (difference 8.9×10^{-16}).

Load measurement. Fig. 2f times a chain of d matrix–vector products ($n = 512$). The linear fit has slope 34.8 μs per step and $R^2 = 0.9985$ in the run shown; across five independent runs R^2 ranged from 0.983 to 0.999, the spread being system timing noise. This supports only the cost model $C(d) = c_0 d$ for matrix chains, not any statement about language-model inference.

VII. Test on Published Accuracy-versus-Compute Data

A. Data and pre-declared protocol

We analysed eight accuracy-versus-compute curves (33 points) whose numbers are printed in tables of two published sources: Tables 4, 10, 11, 18 and 21 of Nie *et al.* [9] (arXiv preprint, May 2026) and Table 6 of Wang *et al.* [10] (ICLR 2023). No value was read from a plot (Table 2). Compute is the output-token budget (C1–C6: Qwen3-8B and DeepSeek-R1-Distill-Llama-8B, with and without visible thinking, on MATH-500 and GSM8K) or the number of sampled reasoning paths (C7–C8: UL2-20B). All tables were extracted twice with a web-reading tool and the two extractions agreed; the token-budget tables come from a preprint that we did not reproduce. Uncertainty is the binomial standard error, or the reported standard deviation over 10 runs for C7–C8.

The test reads the error as $\varepsilon = a_\infty - a$ for accuracy a, a linear read-out of $\lVert E\rVert$ that is an assumption of the test. Then ε is strictly decreasing and convex exactly when a is increasing and concave. The protocol was fixed before any fit was run: **P1** accuracy is non-decreasing in compute (Theorem 1); **P2** accuracy is concave in compute within noise, i.e. the largest z-score of a slope increase between consecutive segments is at most 2 (Theorem 4b; standard errors treated as independent, which is conservative); **P3** on curves with at least four points, compare three 3-parameter models by leave-one-out (LOO) RMSE: SPEC $a = a_\infty - A e^{-g\tau}$ (one spectral mode, $g = \alpha g_1$), POW $a = a_\infty - c\tau^{-b}$ and LOG $a = a_\infty/(1 + (\tau_{50}/\tau)^s)$, with $a_\infty \le 100$. Nonlinear parameters were found by exhaustive grid search.

B. Results

Fig. 4 shows the data and fits; Fig. 5 is the unedited program output, from which every number below is taken.

P1. All 8 curves are monotone.

P2. 6 of 8 curves are concave within noise. Two fail: C2 (think mode, MATH-500; $z = 2.25$, marginal) and C3 (no-think, GSM8K; $z = 3.67$), where the slope rises from 281 to 664 points per 1k tokens between the 32–64 and 64–128 token segments. Table 4 of [9] reports early-stop rates of 0.0% and 2.0% at budgets 32 and 64 (43.5% at 128), i.e. outputs are truncated before an answer exists. For C2, Table 18 of [9] reports 0.0% early stops at budgets 256 and 512. This is consistent with an onset phase in which no answer is available, which the relaxation model $E(\tau) = e^{-\alpha G \tau} E_0$ does not contain (it assumes an answer at every τ); we did not test this explanation directly.

P3. Mean LOO RMSE is 6.08 (SPEC), 17.43 (POW) and 2.32 (LOG) accuracy points, against mean standard errors of 0.88–2.18 points. LOG has the lowest LOO error on 3 curves (C1–C3), SPEC on 1 (C8) and POW on 1 (C7). On the sampling curves C7–C8 SPEC is competitive (LOO 2.10 and 0.39 against

Table 2. Published accuracy-versus-compute curves analysed (values copied from the cited tables; tok = output-token budget).

ID	Model, benchmark (source table)	n	Compute	Accuracy (%)
C1	Qwen3-8B no-think, MATH-500 [9], Tab. 21	500	128, 256, 512, 1024, 2048 tok	4.6, 16.6, 40.6, 59.8, 64.4
C2	Qwen3-8B think, MATH-500 [9], Tab. 18	500	256, 512, 1024, 2048 tok	4.2, 6.2, 18.0, 44.0
C3	Qwen3-8B no-think, GSM8K subset [9], Tab. 4	200	32, 64, 128, 256, 512 tok	3.0, 12.0, 54.5, 89.0, 94.0
C4	Qwen3-8B think, GSM8K subset [9], Tab. 4	200	128, 256, 512 tok	2.0, 22.0, 66.5
C5	R1-Distill-Llama-8B, MATH-500 [9], Tab. 11	500	1024, 2048, 4096 tok	28.4, 40.2, 49.0
C6	R1-Distill-Llama-8B, GSM8K [9], Tab. 10	1319	256, 512, 1024 tok	19.9, 59.6, 63.8
C7	UL2-20B self-consistency, AQuA [10], Tab. 6	10 runs	1, 5, 10, 20, 40 paths	19.7, 24.9, 25.3, 26.7, 26.9
C8	UL2-20B self-consistency, MultiArith [10], Tab. 6	10 runs	1, 5, 10, 20, 40 paths	9.5, 11.3, 12.3, 13.7, 14.7

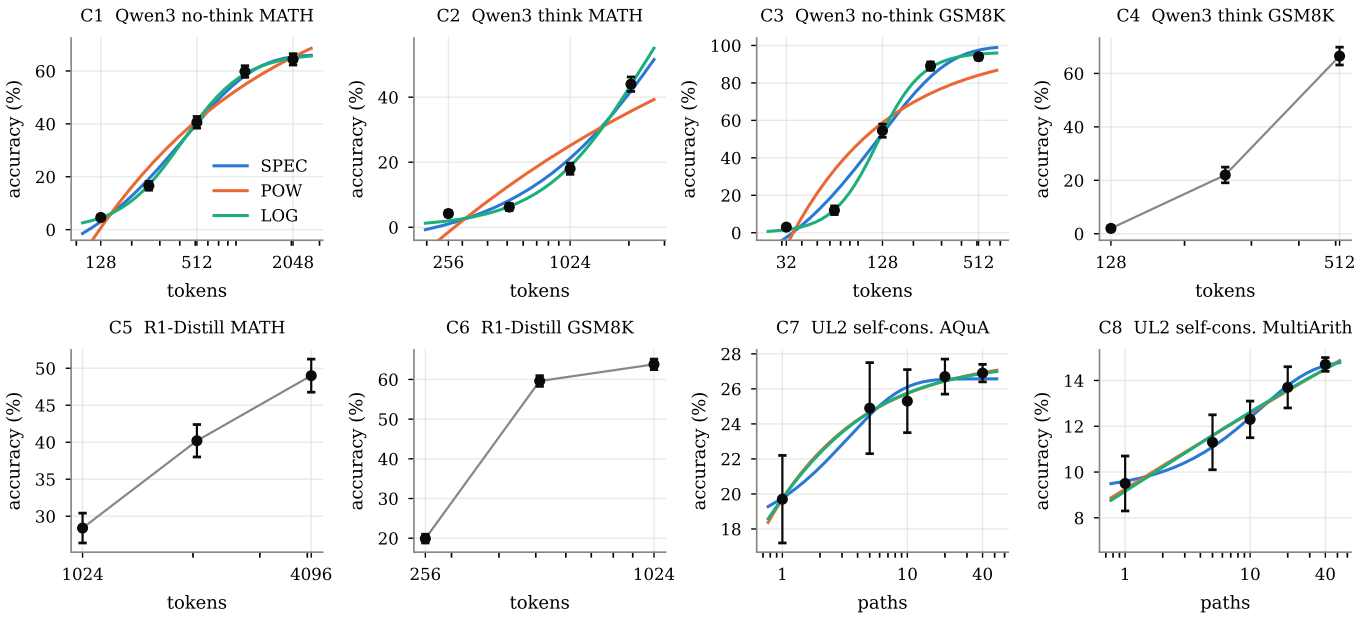


Figure 4. Published curves (black, with standard errors) and the three fitted models: SPEC (single-mode spectral contraction), POW (power law), LOG (logistic in ln compute). Curves with three points (C4–C6) are only plotted. Compute axes are logarithmic.

1.34 and 0.56 for POW). On the token-budget curves with an onset (C1–C3) SPEC is worse than LOG (LOO 5.38, 7.17, 15.36 against 1.64, 3.65, 1.71), and POW is worst because it cannot produce an onset. The fitted SPEC e-folding compute is 0.432 and 0.122 ktok for C1 and C3 and 3.35 and 11.9 paths for C7 and C8; for C2 the fit reaches the bound $a_\infty = 100$, so it is an extrapolation.

Reading. The data support the qualitative laws—accuracy rises with compute (8/8) with diminishing returns in most curves (6/8)—but not the claim that a single relaxation mode describes token-budget curves that contain an onset threshold. A dead-time extension, in which the error stays at $\|E_0\|$ until an onset compute τ_0, is a natural next model; it was not tested here. With 4–5 points per curve and three parameters per model, we make no significance claims, and the evidence covers two math benchmarks, small and medium models, and one preprint.

VIII. Discussion and Limitations

(1) The model is linear with symmetric G; real AI systems are nonlinear and their update matrices need not be symmetric or contractive. (2) Theorems 1–4 are statements about the model. The only empirical evidence is eight published curves from two sources (Section VII); no new measurement of an AI system was made, and the laws remain *hypotheses* for real systems. (3) Falsifiable predictions follow: error versus compute should be a sum of exponentials, with log-error becoming linear in compute at rate αg_1, and an optimal stopping compute where the marginal error reduction equals the latency price. Section VII tested monotonicity and diminishing returns; the optimal-stopping prediction needs a measured latency price and remains untested. Non-monotone behaviour ("overthinking") would require effects outside the model, such as Euler-type overshoot (Proposition 1) or nonlinearity. (4) The mathematical tools are classical; the contribution is the formulation, the corrected analysis and the verification. (5) Cost per step is assumed constant.

IX. Conclusion

A single linear-algebra model, $E(\tau) = e^{-\alpha G \tau} E(0)$, yields the three laws as theorems: error decreases strictly with time per step, decays geometrically with reasoning depth at linear cost, and trades off against speed with a unique optimal latency. The first-order form of the model is stable only for $T < 2/(\alpha g_{\max})$

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MB-Spectral Contraction Theory: test on published accuracy-vs-compute data
sources: NIE = arXiv:2605.07686 (Tabs 4,10,11,18,21); WANG = ICLR 2023 (Tab 6)
curves: 8    points: 33

P1  monotone non-decreasing accuracy (Theorem 1)
  C1 Qwen3-8B no-think MATH-500         PASS
  C2 Qwen3-8B think MATH-500            PASS
  C3 Qwen3-8B no-think GSM8K            PASS
  C4 Qwen3-8B think GSM8K               PASS
  C5 R1-Distill-Llama-8B MATH-500       PASS
  C6 R1-Distill-Llama-8B GSM8K          PASS
  C7 UL2-20B self-consist. AQuA         PASS
  C8 UL2-20B self-consist. MultiArith   PASS
  -> 8/8 curves monotone

P2  diminishing returns: accuracy concave (Theorem 4b); FAIL if max z > 2
    slopes in points per 1k tokens (tok) or per path; z = slope rise / s.e.
  C1 slopes   93.75   93.75   37.50    4.49   max z =  -0.00  PASS
  C2 slopes    7.81   23.05   25.39   max z =   2.25  FAIL
  C3 slopes  281.25  664.06  269.53   19.53   max z =   3.67  FAIL
  C4 slopes  156.25  173.83   max z =   0.59  PASS
  C5 slopes   11.52    4.30   max z =  -2.20  PASS
  C6 slopes  155.08    8.20   max z = -18.97  PASS
  C7 slopes    1.30    0.08    0.14    0.01   max z =   0.09  PASS
  C8 slopes    0.45    0.20    0.14    0.05   max z =  -0.19  PASS
  -> 6/8 curves concave within noise

P3  model comparison, curves with >= 4 points (RMSE in accuracy points; mean s.e. for scale)
    I = in-sample, L = leave-one-out, X = error predicting the last point from the others
  id  n  mean-se |  I:SPEC   POW   LOG |  L:SPEC   POW   LOG |  X:SPEC   POW   LOG
  C1 5    1.83  |   1.61   4.26   0.52  |   5.38  16.37   1.64  |   6.74   0.94   2.45
  C2 4    1.48  |   2.70   7.00   1.19  |   7.17  18.24   3.65  |  10.77  21.69   6.63
  C3 5    2.18  |   5.65  13.38   1.08  |  15.36  50.66   1.71  |   4.04  12.48   1.94
  C7 5    1.68  |   0.43   0.25   0.26  |   2.10   1.34   4.03  |   0.66   0.08   0.06
  C8 5    0.88  |   0.12   0.24   0.27  |   0.39   0.56   0.55  |   0.44   0.38   0.48

  mean leave-one-out RMSE:  SPEC 6.08   POW 17.43   LOG 2.32
  curves where model has the lowest leave-one-out RMSE:  SPEC 1   POW 1   LOG 3

SPEC parameters (a_inf in %, g in 1/ktok or 1/path; 1/g = e-folding compute)
  C1  a_inf =   66.1   g =   2.315   1/g =    0.432 ktok
  C2  a_inf =  100.0   g =   0.297   1/g =    3.364 ktok
  C3  a_inf =   99.5   g =   8.213   1/g =    0.122 ktok
  C7  a_inf =   26.6   g =   0.298   1/g =    3.351 path
  C8  a_inf =   14.9   g =   0.084   1/g =   11.938 path

saved: fig_published.pdf, published_data.csv
```

Figure 5. Verbatim output of `fit_published.py` (unedited).

and has optimum $T^* = 2/(\alpha(g_{\min} + g_{\max}))$. Verification on 5000 random spectra found no violations. On eight published accuracy-versus-compute curves the qualitative laws hold in most cases (monotone 8/8, diminishing returns 6/8), but a single relaxation mode does not describe token-budget curves with an onset threshold. The next steps are an onset (dead-time) extension, a multi-mode fit on denser curves, and a measured latency price to test the equilibrium law.

Data and Code Availability

The verification script `sim.py` (which also generates the figures and numbers of Sections III–VI) and `fit_published.py` with `published_data.csv` (Section VII) accompany this manuscript. Related project code: [11].

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